

## Overview

This study presents the **first publicly available dataset for Pakistan Sign Language (PSL)**, focusing on Urdu alphabet hand configurations. It also introduces a **machine learning-based Android application** that uses this dataset to automatically recognize and translate hand signs into Urdu alphabets. The overarching goal is to bridge the communication gap between deaf and non-deaf communities in Pakistan.

## Methodology

### Data Collection

- **Data Type:** Static images of hand configurations
- **Alphabet Coverage:** 37 Urdu letters
- **Images per Class:** 40
- **Total Dataset Size:** 1,480 images
- **Acquisition Device:** Webcam
- **Variability:** Each sign captured in multiple orientations and hand/finger configurations
- **Format:** Raw and labeled images; annotations stored in XML
- **Dataset Access:** Mendeley Dataset

### Sign Recognition Pipeline

1. **Segmentation:**
  - Used **Hue-Saturation-Intensity (HSI)** color model for robust hand-background separation.
  - Addressed challenges with lighting and color sensitivity by thresholding on hue and saturation.
  - RGB achromatic decomposition used to aid in detecting white-colored hands.

## 2. Detection:

- Utilized shape signatures and **Fast Fourier Transform (FFT)** to ensure invariance to rotation and scale.
- Implemented **homography transformation** for shape localization.

## 3. Recognition:

- **Support Vector Machine (SVM)** was employed using the **Radial Basis Function (RBF)** kernel.
- Used one-vs-all strategy for multi-class classification.
- Hand blobs were normalized and converted to feature vectors for training.

## Mobile Application

- **Platform:** Android
- **Framework:** OpenCV with integrated SVM model
- **Workflow:**
  - Capture image of hand sign using phone camera.
  - Segment and classify the sign using SVM.
  - Display recognized Urdu letter.

## Results

- **Recognition Accuracy:** 80–90% under varying light conditions
- **Metrics Used:**
  - Accuracy
  - Precision
  - Recall
  - F1 score

| Metric    | Formula                                           |
|-----------|---------------------------------------------------|
| Accuracy  | $(TP + TN) / (TP + TN + FP + FN)$                 |
| Precision | $TP / (TP + FP)$                                  |
| Recall    | $TP / (TP + FN)$                                  |
| F1 Score  | $2 * (Precision * Recall) / (Precision + Recall)$ |

- **GUI:** A simple and user-friendly app interface displaying recognized signs in real time.

## Conclusion

This work lays the foundation for PSL-based communication tools by offering a curated dataset and a real-time mobile app. The approach demonstrates that even with simple static images and traditional ML models like SVM, meaningful and practical tools for accessibility can be created. While the system shows good accuracy, there's room for enhancing robustness under diverse real-world conditions.